Landon P. Doughty

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EDUCATION

Rice University - B.S. Electrical Engineering, **GPA: 4.0** **Expected May 2029**
Trustee Distinguished Scholar & **President's Honor Roll**

Relevant Coursework: Digital Logic Design, Random Signals, Physical Electronics I, Fundamentals of Electrical Engineering, Signals, Systems, & Transforms, Fundamentals of Computer Engineering, Analog Circuits Laboratory, Introduction to Engineering Design, Computational Thinking, Linear Algebra, Multivariable Calculus, Wearable Biosensors. Intro to O.R. and Optimization.

RELEVANT EXPERIENCE

Fundamentals of Electrical Engineering Peer Educator and Teaching Assistant **August 2026 - Present**

- Lead weekly office hours and drop-in sessions, providing one-on-one and group support on fundamental concepts.
- Assist students in understanding complex circuit theory and signal processing through clear explanations and guided practice.
- Collaborate with course instructors to identify common student challenges and develop supplementary materials.

Rice Integrated Systems and Electromagnetics Lab - FPGA design and ASIC test intern **May 2026 – Present**

- Architect a custom FPGA data communication system for interface with a custom Neural Interface ASIC.
- Designed a low-power, space constrained PCB, integrating an FPGA, power management, high speed USB, and more.
- Tested and integrated a custom UWB ASIC for transcranial neural interface data transmission.

Rice ECLIPSE - Active Flight Stabilization Lead **August 2025 – Present**

- Lead a cross-functional team in the development of active flight stability systems using controlled fins for high-power rocketry.
- Spearhead end-to-end design of custom flight control software and PCBs, executing iterative testing cycles to ensure flight performance and safety.
- Coordinate semiweekly design reviews and group testing to validate engineering decisions and project timelines.

LEADERSHIP

Hanszen Residential College Beer Bike Team Captain **May 2026 – Present**

- Coordinate logistics, training schedules, and safety protocols for a collegiate racing team, ensuring success in high stakes events.
- Oversee complex mechanical maintenance and inventory for team equipment and bikes, maintaining equipment readiness.
- Serve as the primary liaison between the team, college leadership, and university administration to manage recruitment and external communications.

Doerr Institute - Leading As a First Year Leadership Cohort **September 2025 – May 2026**

- Engaged in intensive leadership development coaching, identifying key growth areas in delegation, conflict resolution, and trust-building.
- Applied feedback from professional leadership developers and peers to improve cross-functional team management and adaptability skills.

SKILLS

Software & Programming: Python, C++, Java, MATLAB/Simulink, LaTeX, HTML/CSS

Electrical Engineering: PCB Design (KiCad), FPGA Design, Circuit Simulation (LTspice), SMT/THT Soldering

Mechanical & Design: SolidWorks, Onshape, FDM 3D Printing & Slicing, Laser Cutting, Shop Tools

Embedded Systems: STM32, ESP32, Raspberry Pi